

ENVIRONMENTAL APPLICATIONS

Maximum Marks: 100

Time allowed: Two hours

1. *Answers to this Paper must be written on the paper provided separately.*
 2. *You will **not** be allowed to write during the first 15 minutes.*
 3. *This time is to be spent in reading the question paper.*
 4. *The time given at the head of this Paper is the time allowed for writing the answers.*
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5. *Attempt **all** questions from **Section A** and **any four** questions from **Section B**.*
 6. *The intended marks for questions or parts of questions are given in brackets[].*

Instruction for the Supervising Examiner

Kindly read aloud the Instructions given above to all the candidates present in the Examination Hall.

This paper consists of 12 printed pages.

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Turn Over

SECTION A (40 Marks)

*(Attempt **all** questions from this **Section**.)*

Question 1

Choose the correct answers to the questions from the given options.

[20]

(Do not copy the questions, write the correct answers only.)

- (i) The Ramsar Convention is an important international treaty to conserve:
- (a) wastelands
 - (b) mountains
 - (c) wetlands
 - (d) ground water
- (ii) The picture given below is of a Panchayat. This is an excellent example of:



- (a) centralised governance
- (b) decentralised governance
- (c) state government
- (d) local government

- (iii) Many software engineers accepted jobs in other countries. What is the impact of their decision on India?
- (a) Recession for India
 - (b) Brain drain for India
 - (c) No impact on India's economy
 - (d) Increase in foreign investment in India
- (iv) Identify the major reasons for desertification from the following:
- 1. Overuse of chemical fertilisers and pesticides
 - 2. Crop rotation
 - 3. Afforestation
 - 4. Overgrazing
- (a) 1 and 4
 - (b) 1 and 2
 - (c) 2 and 3
 - (d) 3 and 4
- (v) Which air pollution control device can reduce Sulphur emission?
- (a) Cyclone separator
 - (b) Electrostatic precipitator
 - (c) Fluidised bed combustion
 - (d) Bag filter

- (vi) The current chairperson of *Dastkar* is:
- (a) Mirra Alfassa
 - (b) Dr. Kousalya Devi
 - (c) Dr. Soundram
 - (d) Laila Tyabji
- (vii) Activated Sludge is a _____ process.
- (a) biochemical
 - (b) physical
 - (c) biological
 - (d) chemical
- (viii) **Assertion (A):** Crop rotation restores soil fertility.
- Reason (R):** Different crops need different nutrients.
- (a) (A) is true and (R) is false.
 - (b) (A) is false and (R) is true.
 - (c) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (d) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (ix) The Taj Trapezium Zone of India has a total area of _____ sq.km.
- (a) 10,400
 - (b) 12,400
 - (c) 14,400
 - (d) 15,400

- (x) Penalties for air pollution in India refer to the_____.
- (a) taxes paid by a person
 - (b) reduction in tax
 - (c) discounts offered after obeying pollution laws
 - (d) punishment incurred on a polluter
- (xi) **Assertion(A):** Indigenous methods of fishing are ecofriendly and maintain a balance in the aquatic ecosystem.
- Reason(R):** Indigenous methods do not disturb the food chain by overfishing or catching non-target fishes.
- (a) (A) is true and (R) is false.
 - (b) (A) is false and (R) is true.
 - (c) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (d) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (xii) Given below is a logo of an international convention. This convention was introduced to _____.



- (a) conserve wetlands
- (b) control transboundary movement of hazardous waste
- (c) reduce greenhouse gases
- (d) protect the ozone layer

- (xiii) Natural Resource Accounting (NRA) deals with _____.
- (a) counting of the number of trees in the forest
 - (b) managing the data on the animals
 - (c) measuring and valuing natural resources
 - (d) calculation of pollution levels
- (xiv) Identify the traditional water conservation systems seen in Rajasthan.
- 1. Johad
 - 2. Eri
 - 3. Kund
 - 4. Percolation pit
- (a) 1 and 2
 - (b) 1 and 3
 - (c) 3 and 4
 - (d) 1 and 4
- (xv) _____ means the use of natural resources which meets the needs of the present generation without compromising the needs of the future generation.
- (a) National development
 - (b) Urban development
 - (c) Sustainable development
 - (d) Rural development

- (xvi) Which of the following statements about *Environmental Impact Assessment (EIA)* is **APPROPRIATE**?
- (a) EIA is a quick method.
 - (b) The cost of EIA process is low.
 - (c) EIA does not require the cooperation of the local people.
 - (d) EIA helps in reducing negative impacts of the project.
- (xvii) Which of the following statements is **TRUE** regarding the role of local people in Joint Forest Management (JFM) in India?
- (a) Local people act as the guardians of the forest.
 - (b) Local people cannot collect forest products.
 - (c) Local people destroy the forest.
 - (d) Local people are not involved in the conservation programme.
- (xviii) Which of the following is an example of indigenous knowledge?
- (a) Modern farming techniques
 - (b) Satellite weather forecasting
 - (c) Genetic engineering of crops
 - (d) Handloom weaving
- (xix) The main gases produced by landfills are:
- (a) Methane and ethane
 - (b) Methane and carbon dioxide
 - (c) Hydrogen sulphide and carbon monoxide
 - (d) Ethane and sulphur dioxide

(xx) Which of the following is **NOT** a modern method of water harvesting?

- (a) Borewells
- (b) Kunds
- (c) Percolation pits
- (d) Recharge trenches

Question 2

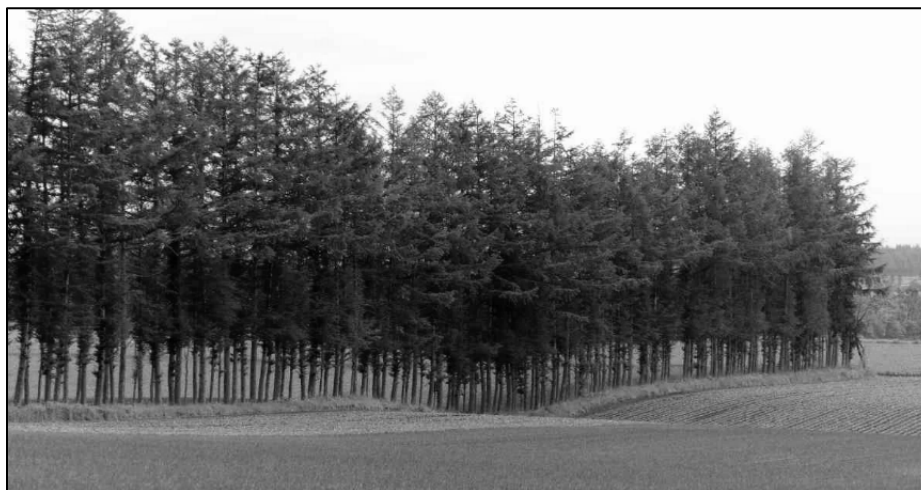
(i) What is meant by *End-of-pipe treatment*? [2]

(ii) *The western coastal region and parts of the northern coastal areas of India are highly vulnerable to soil erosion.* [2]

In this context, discuss *any two* consequences of soil erosion.

(iii) *Dry Compost Toilets are an alternative to the existing sewage system.* [3]
State *any three* benefits of DCT.

(iv) Observe the picture given below and name the *soil conservation strategy*. [3]
State *two* reasons why this strategy is useful to the farmers.



Question 3

- (i) Mention *any two* objectives of Gandhigram. [2]
- (ii) *Biodegradable waste is beneficial for the environment.* [2]
Give *two* reasons to support this statement.
- (iii) State *three* basic steps involved in effluent treatment plant. [3]
- (iv) Mention *any three* initiatives you can take to conserve water resources in your locality. [3]

SECTION B (60 Marks)

(Answer *any four* questions from this *Section*.)

Question 4

- (i) Observe the picture given below and answer the questions that follow: [5]



- (a) Identify the waste disposal method shown in the figure above.
 - (b) Explain *any four* best practices to tackle the issue mentioned by you in (a).
- (ii) Explain *any five* alternatives to fuelwood to save the environment. [5]
- (iii) Give *any five* reasons for the treatment of wastewater before it is discharged in the environment. [5]

Question 5

- (i) *Air pollution control devices help to reduce air pollution effectively.* [5]

In this context, describe the working principle of wet scrubbers.

- (ii) Discuss *any five* reasons to justify that globalisation is affecting our natural resources. [5]

- (iii) Given below are images of the monuments falling under *Taj Trapezium Zone (TTZ)*. [5]



Discuss *any five* rulings given by the Supreme Court of India *to protect our monuments* under TTZ.

Question 6

- (i) What is rooftop rainwater harvesting? Discuss *any three* reasons why we should conserve rainwater. [5]

- (ii) *Technologies are creating a great impact in restoring our natural resources.* [5]
Give a brief account of *five* such technologies which help to save our natural resources.

- (iii) Discuss *any five* strategies to conserve soil to prevent generation of waste land. [5]

Question 7

- (i) Suggest *any five* modern techniques to improve traffic management in India. [5]
- (ii) *Conducting Environmental Impact Assessment (EIA) is necessary for the big projects in India.* [5]

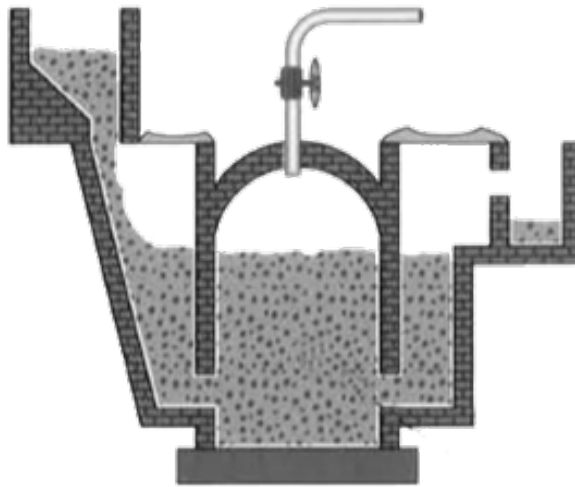
In this context, explain EIA. Mention *any two* advantages and *any two* disadvantages of EIA.

- (iii) *A watershed includes both the physical and the biological features of the land and the water bodies it contains.* [5]

Briefly explain *five* objectives of watershed management.

Question 8

- (i) Observe the structure shown in the diagram given below. [5]



- (a) Name the structure shown above.
- (b) Describe the working of the structure mentioned by you in (a) above.
- (c) List *two* advantages of this structure.

- (ii) *Mahatma Gandhi promoted Khadi as a symbol of the Swadeshi Movement in India. As a young citizen of India, write any five reasons for promoting Khadi.* [5]
- (iii) *Discuss any five objectives of sustainable development.* [5]

Question 9

- (i) *Many international, national and local NGOs are working in different areas to improve the condition of the environment.* [5]
State *five* measures taken by NGOs towards environmental protection.
- (ii) *The global demand for remote sensing satellites has increased substantially overtime.* [5]
Describe *any five* reasons why remote sensing satellites are considered such an important tool.
- (iii) *Panchayati Raj system was introduced to empower the villagers of rural India.* [5]
Discuss *any five* functions of Panchayat.